

# Hatice Merve Vural

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## EDUCATION

- Koc University** Istanbul, Turkey  
*M.Sc. in Computer Science and Engineering, CGPA: 3.91* Sep 2024 - current
- Sabanci University** Istanbul, Turkey  
*B.S. in Computer Science and Engineering, CGPA: 3.44; Minor in Mathematics, CGPA: 3.82* Aug 2019 - 2024
- Malardalen University** Vasteras, Sweden  
*Erasmus exchange Program in School of Innovation, Design and Engineering* Jan 2023 - June 2023

## AWARDS AND SCHOLARSHIPS

- Koc University Graduate Scholarship** (31,000 TL monthly stipend, with meal card and free dormitory housing)
- Full Scholarship from Sabanci University** (covering tuition and free dormitory housing)
- Received 1389th rank among 2M high school graduates** in University Entrance Examinations.

## RESEARCH INTEREST

- Humor understanding, visual perception, and reasoning in multimodal large language models (MLLMs).
- Developing neural network algorithms for domain adaptation and generalization.
- Researching sampling methods for probabilistic modeling and inference.
- Investigating Monte Carlo methods and their efficiency in Bayesian statistics.

## RESEARCH ARTICLES

- Hatice Merve Vural, Doga Kukul, Ege Erdem Ozlu, Demir Ekin Arikan, Bob Mankoff, Erkut Erdem, Aykut Erdem, "Learning to Think Like a Cartoon Captionist: Incongruity-Resolution Supervision for Multimodal Humor Understanding", arXiv:2604.15210, 2026. (under review)

## EXPERIENCE

- Koc University** KUIS AI Lab  
*Research* Feb 2025 - current
  - Enhancing the humor understanding, visual perception, and reasoning abilities of multimodal large language models (MLLMs).
  - Post-training MLLMs to transfer their capabilities to underexplored domains.
- Sabanci University** Computer Science Laboratory  
*Graduation Project* Oct 2023 - June 2024
  - Aims to develop a novel domain generalization algorithms to increase accuracies of state-of-the-art algorithms.
  - Evaluate existing algorithms for domain generalization.
  - Simulate varied data scenarios to address limitations of current approaches.
- University of Toronto**  
*Undergraduate Research Internship* June 2023 - Oct 2023
  - Aimed to implement different Monte Carlo Markov Chain algorithms and compare their convergence rate.
  - Implemented Metropolis-Hastings and Langevin algorithm in Python. Analyzed their plots.
  - Calculated and plotted theoretical convergence rates with respect to Wasserstein-2 distances.

## TEACHING EXPERIENCE

- Koc University** Introduction to Computer Science and Programming - Comp100  
*Teaching Assistant* Feb 2025 - current
  - Assisting students with programming concepts in Python.
  - Preparing and conducting weekly pre-lab and in-lab sessions, as well as problem sets.
- Koc University** Discrete Mathematics for CS and Engineering - Comp106  
*Teaching Assistant* Sep 2024 - Feb 2025
  - Assisting in discrete mathematics concepts for engineering.
  - Conducting weekly problem solving sessions and quizzes.

- Sabancı University** Introduction to Probability-Math203  
*Teaching Assistant* Feb 2024 - June 2024
  - Assisting in probability concepts.
  - Solving questions and helping students for their probability problems in weekly recitation classes.
- Sabancı University** Computer Science Laboratory  
*Undergraduate Research Assistant* Oct 2022 - Jan 2023
  - Aimed to develop a generic deep learning algorithm for detection of different types of bacteria using the fluorescent and phase images of cells.
  - Conducted image segmentation with imageJ.
- Sabancı University** Advanced Programming-CS204  
*Teaching Assistant* Oct 2022 - Jan 2023
  - Assisted in advanced programming in C++.
  - Helped students for object oriented programming problems in weekly recitation classes.
- Sabancı University** Collaboration Space  
*Technical Assistant* March 2022 - July 2022
  - Conducted projects with arduino and raspberry Pi.
  - Helped students for arduino projects.
- Sabancı University** Introduction to Computing-CS201  
*Teaching Assistant* Sep 2021 - June 2022
  - Assisted in teaching programming in C++.
  - Helped students for programming problems in weekly recitation classes.

## SKILLS

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- Languages:** Turkish (Native), English (TOEFL IBT: 110/120 (5.5/6.0))
- Programming:** Python, C++, Dart, SQL, php, openGl, pyTorch, TensorFlow
- Platforms:** macOS, Web, Windows, Arduino, Raspberry Pi

## PROJECTS

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- Domain Generalization**  
<https://github.com/mervehmv/DomainGeneralization>
  - developed a novel domain generalization algorithm to increase accuracies of state-of-the-art algorithms.
- Monte Carlo Markov Chain**  
<https://github.com/mervehmv/MCMC>
  - implemented Monte Carlo Markov Chain algorithms (Metropolis-Hastings, Unadjusted Langevin, and Metropolis adjusted Langevin Algorithm).
  - proved theoretical convergence rates with Wasserstein-2 distances and plotting the theoretical rates.
- Arduino 'Truth or Dare' Project**  
<https://github.com/mervehmv/truthAndDare>
  - Aimed to assign a questioner and a player who chooses truth or dare.
  - Prepared a box with 8 leds and a button and an arduino inside.
  - Programmed arduino to blink lights in order, whenever pressed the button it stops in one led to choose a player.
- suSell : Online Shopping App**  
<https://github.com/mervehmv/suSell>
  - Aimed an easier way for shopping inside the campus among students
  - Prepared with several categories with location to facilitate shopping.
- Bitcoin Price Prediction Project**  
<https://github.com/mervehmv/bitcoin>
  - Aimed to indicate trends between volume and opening price, volume and closing price, volume and number of tweets in a day that contains Bitcoin.
  - Predicted volume, opening and closing prices for 30 days.

- **Lab monitoring with IoT**  
<https://github.com/mervehmv/labmonitoringwithiot>
  - Aimed to measure temperature with raspberry Pi and a temperature sensor.
  - Tracked temperatures online from the internet.
- **musicLibrary**  
<https://github.com/mervehmv/musicLibrary>
  - Created a database of songs, singers and albums with sections of search, filter, add, and delete.

## VOLUNTEER EXPERIENCE

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- **Volunteer** Civic Involvement Projects - CIP101  
*Primary School*
  - Volunteered for helping primary school students to understand important concepts such as climate change, human body, etc.

## INTEREST

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- **Music, Sport**
  - Playing electronic guitar and drums.
  - Taekwondo, MMA, hiking